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Research Interest

My research interests broadly include the domains of cloud computing and network systems security, with a particular focus on improving performance and efficiency. Within the realm of cloud computing, my focus is on exploring innovative approaches to safeguard sensitive data, mitigate emerging threats, and ensure the privacy and integrity of cloud environments. I am particularly interested in developing resource optimization techniques, robust security mechanisms, and access control strategies tailored to the unique challenges faced in cloud computing. Additionally, I am intrigued by the integration of emerging technologies such as serverless architectures, as they offer to optimize the overall performance of cloud-based services. In the field of network system security, my research aims to develop robust security mechanisms, threat detection, and mitigation strategies to safeguard critical infrastructures. Leveraging machine learning, I aim to address emerging vulnerabilities and ensure the confidentiality, integrity, and availability of data and services.

Education

Ph.D., School of Electrical Engineering (Advisor: Dr. Seungwon Shin) *March 2020 - August 2026*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea

M.S., School of Electrical Engineering (Advisor: Dr. Seungwon Shin) *March 2018 - February 2020*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea

B.S., School of Electrical Engineering *March 2013 - February 2018*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea

Publications

1. **Kang Haney**, and Seungwon Shin. "Verikube: Automatic and Efficient Verification for Container Network Policies." IEICE TRANSACTIONS on Information and Systems 105.12 (2022): 2131-2134.

2. **Kang Haney**, Jinwoo Kim, and Seungwon Shin. "Minicon: Automatic enforcement of a minimal capability set for security-enhanced containers." 2021 IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS). IEEE, 2021.

3. **Kang Haney**, Eduard Marin, Myoungsung You, Diego Perino, Seungwon Shin, Jinwoo Kim. "BEACON: Automatic container policy generation using environment-aware dynamic analysis." Computers & Security (2025): 104789.

4. **Kang Haney**, Myoungsung You, Seungwon Shin, Jinwoo Kim. "PROMPT2SECCOMP: Semantics-Aware Automated Security Policy Generation for Containers using Large Language Models." Under Review.

Experience

KAIST *Daejeon, South Korea*
Teaching Assistant Spring 2018 - 2022

- EE477 Database and Big Data Systems
- EE485(A) Special Topics in Electronic Engineering <Software development environment and tools practice>
- EE209 Programming Structure for Electrical Engineering
- EE309 Advanced Programming Techniques for Electrical Engineering
- EE513 Network Systems and Security
- EE488(A) Special Topics in Electrical Engineering <Programming for Electrical engineering>

Undergraduate Research Intern Spring 2017 - 2018

- Software Define Networking Security Framework
- Network Function Virtualization
- Network System Security
- Advisor: Dr. Seungwon Shin

SK Hynix *Cheongju, South Korea*
Undergraduate Intern Summer 2016

Relevant Coursework

Computer System

Computer Architecture, Operating System, Advanced Parallel Architecture

Network System

Network Programming, Network Systems and Security, Distributed Systems

Information Security

Information Security

Machine Learning

Database and Bigdata System, Foundation of Big Data Analytics, Deep Reinforcement Learning

Skills

Languages: C, C++, Python, Java, MATLAB
Databases: MySQL, MongoDB
Cloud: Docker, Kubernetes, OpenFaaS